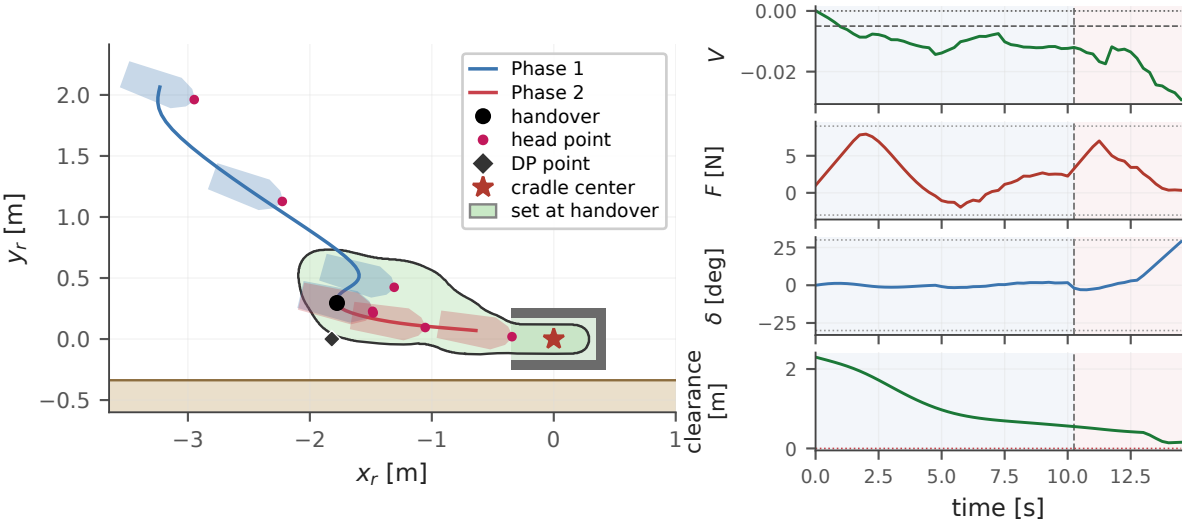
Phase 1
Phase 2
handover
head point
DP point
cradle center
set at handover
$y_r$ [m]
$x_r$ [m]
2.0
1.5
1.0
0.5
0.0
−0.5
−3
−2
−1
0
1
$V$
0.00
−0.02
$F$ [N]
5
0
$\delta$ [deg]
25
0
−25
clearance [m]
2
0
0.0
2.5
5.0
7.5
10.0
12.5
time [s]